Java Assignment

1. Boat

I used a frequency array here so that I can keep count of each number. For every possible sum, I check which two values can make that sum. If the values are different, the smaller frequency decides how many pairs can be made. If they are the same, I divide the frequency by 2.

```
import java.util.*;

public class Main{
    static int maxTeams(int[] arr,int[] freq){
        int result=0;

        for(int sum=2;sum<=2*arr.length;sum++){
            int teams=0;

            for(int x=1;x<=arr.length;x++){
                int y=sum-x;

                if(y<=0||y>arr.length)
                    continue;

                if(x>y)
                    continue;

                if(x<y)
                    teams+=Math.min(freq[x],freq[y]);
                else
                    teams+=freq[x]/2;
            }

            result=Math.max(result,teams);
        }

        return result;
    }

    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        int t=sc.nextInt();

        while(t-->0){
            int n=sc.nextInt();
            int[] arr=new int[n];
            int[] freq=new int[n+1];

            for(int i=0;i<n;i++){
                arr[i]=sc.nextInt();
                freq[arr[i]]++;
            }

            System.out.println(maxTeams(arr,freq));
        }
    }
}
```

2. Three parts of array

I kept one pointer on the left side and one on the right side. The sums of both sides are stored separately. When the sums are equal, I store that sum. Otherwise, I move the pointer from the side having the smaller sum. The loop continues until both pointers meet.

```
import java.util.*;

public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        int n=sc.nextInt();
        int[] arr=new int[n];

        for(int i=0;i<n;i++)
            arr[i]=sc.nextInt();

        int l=0;
        int r=n-1;

        long left=arr[l];
        long right=arr[r];
        long ans=0;

        while(l<r){
            if(left==right){
                ans=left;
                l++;
                left+=arr[l];
            }
            else if(left<right){
                l++;
                left+=arr[l];
            }
            else{
                r--;
                right+=arr[r];
            }
        }

        System.out.println(ans);
    }
}
```